00924

Sri Lanka Institute of Information Technology

B.Sc. Honours Degree in Information Technology

Specialized in Information Technology

Final Examination
Year 3, Semester 2 (2023)

IT3080 – Data Science and Analytics

Duration: 2 Hours

November 2023

Instructions to Candidates:

- This paper is preceded by 10 minutes reading period. The supervisor will indicate when answering may commence.
- This paper has 4 questions.
- Answer all questions in the booklet given.
- The total marks for the paper is 100.
- This paper contains 5 pages, including the cover page.
- Calculators are allowed. Any other electronic devices capable of storing and retrieving text, including mobile phones are not allowed.

Question 1

30 Marks

a. Name two techniques that could be used to analyze data. (2 marks)

b. Briefly explain how the data analysis is useful to obtain better business prospect for business organizations.

(3 marks)

c. Answer the following questions in relation to data pre-processing.

i. What is meant by dimensionality reduction? (2 marks)

ii. Explain two types of dimensionality reduction techniques using suitable examples. (6 marks)

d. i. Consider the following dataset. Assume the minimum support is 50%. Identify the frequent 2 - item sets within the set of transactions using the A priori property. (5 marks)

1000	Jacket, Shoes
1001	Boots, Coat, Ski Pants
1002	Jacket, Shoes, Ski Pants
1003	Coat, Shoes, Sweater
1004	Boots, Ski Pants
1005	Shoes, Sweater
1006	Jacket, Ski Pants
1007	Coat, Shoes

ii. Assuming that minimum confidence is 40% determine whether *Ski Pants -> Shoes* is a strong association rule or not.

(3 marks)

e. Briefly explain the density-based clustering method.

(3 marks)

f. Consider the following 4 observations in a 2D space. The distance between observations calculated using Euclidean distance is given.

	A	B	C	D
A	0			
B	6	0		
C	4	3	0	
D	5	2	15	0

Use complete link agglomerative clustering to group the data. Show how the clustering is performed step by step. Show the dendrograms also in your answer. (6 marks)

Question 2

30 Marks

a. Name 2 techniques that can be used in data classification. (2 marks)

b. Consider the following dataset.

High	High	High	No
High	Low	High	No
Low	Low	High	Yes
Low	High	High	No
High	High	Low	Yes
Low	Low	Low	Yes
Low	High	Low	Yes
High	Low	High	No

i. Consider the *Income* as the root node and construct the binary decision tree to predict whether a person would have a credit risk or not under the given conditions based on the dataset. Show the steps you have followed to construct the tree in your answer.

(12 marks)

i. Using the constructed decision tree predict whether a person with a high income with low savings and high assets would have a credit risk or not.

(2 marks)

c. What is multiple regression and when is it used?

(3 marks)

d. Consider the following output obtained by performing regression analysis to predict the 'VO_2max' (an indicator of fitness and health) of persons based on the age, weight, heart rate and gender.

The model resulted adjusted R-squared of 0.877 and F statistic of 32.393 with a p- value of 0.0000.

	coef	std err	t	p>|t|
const	87.830	6.385	13.756	.000
age	-0.165	0.063	-2.633	.010
weight	-0.385	0.043	-8.877	.000
heart_rate	-0.118	0.032	-3.667	.000
gender	13.208	1.344	9.824	.074

i. Write the multiple linear regression equation for the model above.

(4 marks)

ii. Predict the VO_2max of a person who is 37 years old, 40 Kg weight with the heart rate 110 bpm.

(3 marks)

e. What is meant by the random or irregular component in a time series? Provide an example where a random or irregular component may be seen in a time series.

(4 marks)

Question 3

25 marks

a. Name 4 layers in the data warehouse architecture.

(4 marks)

b. What is the usage of slowly changing dimension? Explain SCD Type 1.

(4 marks)

c. Management of a car rental company expects to identify how the rentals are performing. For the above purpose they require to answer the following questions.

- How many unrented cars are there in a particular month from each brand?
- How much income is obtained by renting cars to customers in a year?
- How many Toyota cars are rented by customers from Europe?
- How many Germans have rented BMW cars during last month?
- How much income is obtained by renting Benz and Nissan cars in the 1st quarter?
- How many males rented Toyota cars in last year?
- How many European females below 30 years rented BMW cars?
- How much income generated by American males above 50 years who rented the car for one week and travelled more than 1000km?

To answer the above queries the data the car rentals have, are expected to be stored in a data warehouse. Propose a star schema for a data warehouse to cater the above requirements. Select attributes as required for the dimension and fact tables.

(12 marks)

d. Briefly explain 2 operations performed on a multidimensional model using examples.

(5 marks)

Question 4 — 15 Marks

a. List 3 applications in Hadoop. (3 marks)

b. What is meant by *Cluster Rebalancing* in Hadoop? Briefly explain it. (4 marks)

c. The Sri Lankan airline tracks the on-time performance of domestic flights operated by large air carriers. Summary information on the day of the week (DayOfWeek), origin airport (Origin), and departure delay (DepDelay) are published in DOT's monthly Air Travel Consumer Report. The following input file contains a sample of data that can be used for the analysis.

DayOfWeek	Origin	DepDelay (mins)
1	23.4	32
1	33.1	40
2	22.4	25
2	25.6	15
3	32.1	8
3	30.0	18
3	28.7	22

d. Draw MapReduce workflow to find the average departure delay time by day of week. Indicate the results in split, map, combine, shuffle/ sort, and reduce phases. Assume that this involves two mappers. (5 marks)

i. What is the <key, value> in the map phase? (1 mark)

ii. What is the <key, value> in the combine phase? (1 mark)

iii. What is the <key, value> in reduce phase? (1 mark)

---END OF QUESTION PAPER—